FNCE 30007 DERIVATIVE SECURITIES

Tutorial Questions

Forward and Futures pricing

Question 1

A one-year long forward contract on a non-dividend paying stock is entered into when the stock price is $40 and the risk-free rate of interest is 10% per annum with continuous compounding.

a) What are the forward price and the initial value of the forward contract?
b) Six months later, the price of the stock is $45 and the risk-free interest rate is still 10%. What are the forward price and the value of the forward contract?

Question 2

A stock is expected to pay a dividend of AUD 1 per share in two months and in five months. The stock price is AUD 50 and the risk-free rate of interest is 8% per annum with continuous compounding for all maturities. An investor has just taken a short position in a six-month forward contract on the stock.

a) What are the forward price and the initial value of the forward contract?
b) Three months later, the price of the stock is AUD 48 and the risk-free rate of interest is still 8% per annum. What are the forward price and the value of the short position in the forward contract?

Question 3

Suppose that the risk-free rate is 10% per annum with continuous compounding and that the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities does this create?

Question 4

The two-month interest rates in Switzerland and the United States are 2% and 5% per annum, respectively, with continuous compounding. The spot price of the Swiss franc is $0.80. The futures price for a contract deliverable in two months is $0.81. What arbitrage opportunities does this create?

Question 5

Suppose that F_1 and F_2 are two futures contracts on the same commodity with times to maturity, t_1 and t_2, where $t_2 > t_1$. Prove that

$$F_2 \leq F_1 e^{r(t_2 - t_1)}$$

where r is the interest rate (assumed constant) and there are no storage costs. For the purposes of this problem, assume that a futures contract is the same as a forward contract.

Question 6

When a known future cash outflow in a foreign currency is hedged by a company using a forward contract, there is no foreign exchange risk. When it is hedged using futures contracts, the marking-to-market process does leave the company exposed to some risk. Explain the nature of this risk. In particular, consider whether the company is better off using a futures contract or a forward contract when:

a) The value of the foreign currency falls rapidly during the life of the contract.
b) The value of the foreign currency rises rapidly during the life of the contract.
c) The value of the foreign currency first rises and then falls back to its initial value.
d) The value of the foreign currency first falls and then rises back to its initial value.

Assume that the forward price equals the futures price.

Question 7

The following table (also in excel file T3_Q7_data.xls) contains data on monthly changes in the spot price and the futures price for a certain commodity. Using the approach illustrated in the S&P 500 hedge ratio example, calculate the minimum variance hedge ratio from the given data.

Spot_Change	Futures_Change
0.5	0.56
0.61	0.63
-0.22	-0.12
-0.35	-0.44
0.79	0.6
0.04	-0.06
0.15	0.01
0.7	0.8
-0.51	-0.56
-0.41	-0.46